

CogTutor: A Cognitive Science-Inspired AI Study Assistant for Promoting Active Learning

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Abstract— AI study tools often provide immediate answers, which may reduce opportunities for active learning. This paper presents CogTutor, a cognitive-science-inspired study assistant that encourages learners to think before receiving the final answer. The prototype incorporates retrieval practice, guided hints, self-explanation, metacognitive reflection, and delayed feedback into a structured learning sequence. CogTutor was implemented as a Streamlit-based proof of concept and functionally evaluated to verify the learning workflow. The project demonstrates how cognitive science principles can be incorporated into AI-assisted educational tools that promote active learning.

Keywords— *cognitive science, active learning, retrieval practice, metacognition, self-explanation, AI tutoring systems, educational technology.*

I. INTRODUCTION

Artificial intelligence has become an increasingly common learning resource for university students. While AI-powered tools can quickly answer questions and explain concepts, they may also encourage students to rely on immediate solutions instead of actively engaging with the learning process. As a result, opportunities for retrieval practice, self-explanation, and metacognitive reflection may be reduced, limiting deeper understanding and long-term retention.

Cognitive science suggests that effective learning requires more than simply receiving correct answers. Strategies such as retrieval practice, guided questioning, self-explanation, and delayed feedback encourage learners to actively construct knowledge and monitor their own understanding. Rather than replacing these processes, AI has the potential to support them when designed with appropriate instructional principles.

This project presents **CogTutor**, a Streamlit-based proof-of-concept AI study assistant that applies cognitive science principles to encourage active learning. Instead of immediately providing complete answers, CogTutor guides learners through a structured sequence consisting of retrieval questions, guided hints, self-explanation, metacognitive reflection, and delayed answer disclosure. The goal is to demonstrate how AI can function as a learning partner that promotes engagement rather than passive answer consumption.

II. RESEARCH QUESTION

A. Research Question

How can an AI study assistant encourage retrieval practice, self-explanation, metacognitive reflection, and appropriate AI use among graduate students?

This research question addresses the TA's feedback by clearly identifying the instructional features being evaluated and the intended learning outcomes.

B. Objectives

- Design and implement a cognitive-science-inspired AI study assistant that promotes active learning.
- Evaluate how guided questioning and self-explanation encourage retrieval practice and metacognitive reflection.
- Examine whether delayed answer disclosure reduces passive reliance on AI while supporting meaningful learning.

III. COGNITIVE SCIENCE BACKGROUND

CogTutor is grounded in cognitive science principles that encourage active learning instead of passive answer consumption. Retrieval practice requires learners to recall information before receiving feedback, which promotes deeper learning and long-term retention [1]. Therefore, CogTutor begins each interaction with a guided question rather than an immediate answer.

The system also incorporates self-explanation and metacognition to encourage learners to explain their reasoning and reflect on their understanding. These strategies help learners monitor their knowledge, identify misconceptions, and regulate their own learning process [2], [3].

Finally, CogTutor draws on Hutchins' theory of distributed cognition, which views cognition as emerging through interactions among people, tools, and external representations rather than within a single individual [4]. In this project, the AI study assistant serves as a cognitive support tool that guides learners through structured reasoning instead of replacing their thinking. Together, these cognitive science principles informed the design of CogTutor's guided questioning, reflection prompts, and delayed answer disclosure.

IV. MODEL AND TOOL DESIGN

CogTutor is a Streamlit-based study assistant designed to promote active learning through retrieval practice, guided hints, self-explanation, metacognitive reflection, and delayed feedback [1], [3]. Instead of giving an immediate answer, the system moves the learner through a five-stage sequence that requires an initial attempt before support is provided.

The prototype includes four predefined learning topics and operates in demonstration mode. After the final explanation, a learning dashboard summarizes topic completion, completed stages, and recommended next steps. The design treats the tool as cognitive support rather than a replacement for learner reasoning, consistent with distributed cognition [4].

The workflow is:

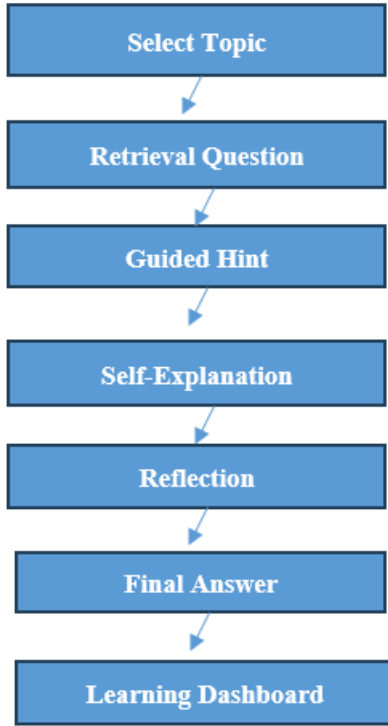


Fig. 1. Overall workflow of the CogTutor learning process.

V. PROTOTYPE IMPLEMENTATION

A proof-of-concept prototype of CogTutor was developed using the Streamlit framework. The interface guides learners through a structured learning sequence based on cognitive science principles, including retrieval practice, guided hints, self-explanation, metacognitive reflection, and delayed feedback. The following figures illustrate the main components of the implemented system.

A. CogTutor Home Interface

The home interface allows learners to select a learning topic and presents the five-stage learning workflow. The sidebar also displays the learner's progress throughout the session.

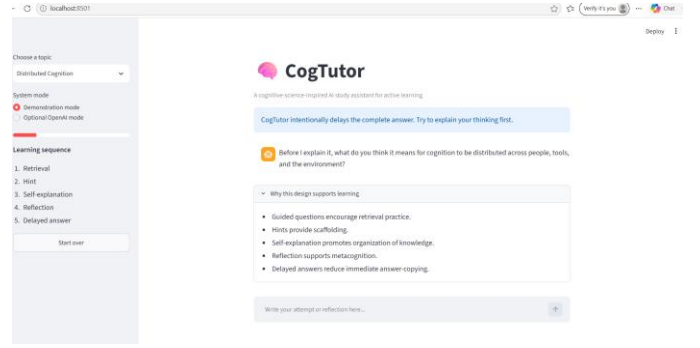


Fig. 2. CogTutor home interface.

B. Retrieval practice stage.

The learning session begins with a retrieval question that encourages learners to recall prior knowledge before receiving instructional support.

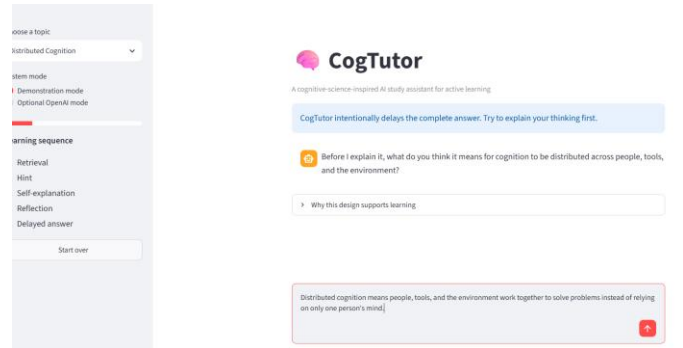


Fig. 3. Student attempts retrieval before receiving assistance.

C. Guided hint stage

After the learner submits an initial response, CogTutor provides a guided hint instead of revealing the complete answer, encouraging continued reasoning.

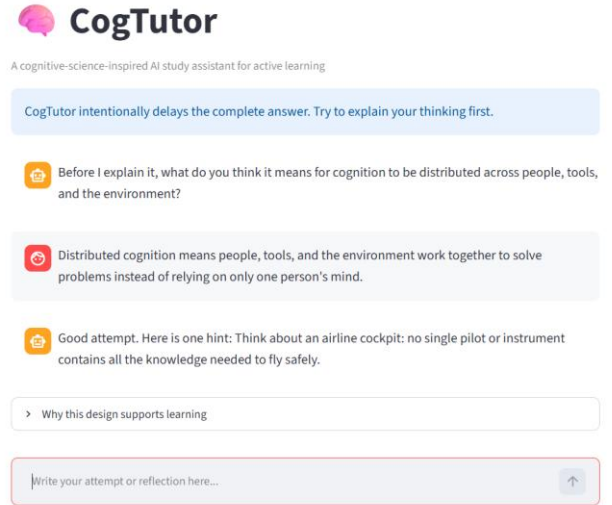


Fig. 4. Incremental hint provided after learner retrieval attempt.

D. Self-explanation stage

Learners are prompted to explain their reasoning in their own words, promoting deeper understanding and knowledge organization [2].

A cognitive-science-inspired AI study assistant for active learning

CogTutor intentionally delays the complete answer. Try to explain your thinking first.

Before I explain it, what do you think it means for cognition to be distributed across people, tools, and the environment?

Distributed cognition means people, tools, and the environment work together to solve problems instead of relying on only one person's mind.

Good attempt. Here is one hint: Think about an airline cockpit: no single pilot or instrument contains all the knowledge needed to fly safely.

> Why this design supports learning

A person and a tool become one cognitive system because information is shared between them to complete a task

Fig. 5. Self-explanation stage encouraging deeper learning.

E. Metacognitive reflection stage

CogTutor asks learners to reflect on their confidence and identify remaining uncertainties, supporting metacognitive monitoring [3].

Good attempt. Here is one hint: Think about an airline cockpit: no single pilot or instrument contains all the knowledge needed to fly safely.

A person and a tool become one cognitive system because information is shared between them to complete a task.

In your own words, explain how a person and an external tool can form one cognitive system.

I am fairly confident, but I still want a better understanding of how cognition is distributed across multiple people.

What part of your explanation feels strongest, and what part are you still unsure about?

> Why this design supports learning

Write your attempt or reflection here...

Fig. 6. Metacognitive reflection encouraging self-monitoring.

F. Delayed answer disclosure

The complete explanation is revealed only after the learner has completed the previous learning activities, encouraging active engagement before feedback [1]

CogTutor

A cognitive-science-inspired AI study assistant for active learning

CogTutor intentionally delays the complete answer. Try to explain your thinking first.

- Before I explain it, what do you think it means for cognition to be distributed across people, tools, and the environment?
- Distributed cognition means that thinking can be shared across people, tools, and the environment.
- Good attempt. Here is one hint: Think about an airline cockpit: no single pilot or instrument contains all the knowledge needed to fly safely.
- For example, pilots and cockpit instruments work together because information is distributed across the whole system.
- In your own words, explain how a person and an external tool can form one cognitive system.

Fig. 7. Delayed explanation shown after active learning activities are completed.

G. Learning dashboard

After completing the learning sequence, CogTutor summarizes learner progress, completed stages, and recommends the next topic. The dashboard provides a summary of learner progress and completion of the cognitive learning sequence.

Choose a topic
Distributed Cognition

System mode
Demonstration mode
Optional Open mode

Learning sequence
1. Retrieval
2. Hint
3. Self-explanation
4. Reflection
5. Delayed answer
Start over

I understand the main idea, but I still want to learn more about how information moves between people and external artifacts.

Distributed cognition is the view that cognitive activity can extend beyond one person's mind. Information may be represented, transformed, and coordinated across people, artifacts, and the environment.

Learning cycle completed

Learning Dashboard

Topic	Stages Completed
Distributed Cognition	5 / 5

Learning Progress: 100%
Status: Completed

> Why this design supports learning

Fig. 8. CogTutor learning dashboard summarizing topic completion, learning progress, completed stages, and status.

VI. PROTOTYPE EVALUATION

CogTutor was evaluated as a proof-of-concept prototype by verifying the implementation of each stage in the cognitive learning workflow. Functional testing confirmed that learners could complete the retrieval, guided hint, self-explanation, reflection, delayed feedback, and learning dashboard stages successfully. In addition, a pilot evaluation questionnaire was developed to support future user studies. The questionnaire includes Likert-scale and open-ended questions to assess learner engagement, retrieval practice, metacognitive reflection, and overall usability.

VII. PRELIMINARY FINDINGS

The completed prototype demonstrates that cognitive science principles can be successfully translated into an interactive educational tool. The implemented workflow supports active learning through structured guidance rather than immediate answer delivery. Although a formal user study was outside the scope of this project, the developed evaluation instrument provides a foundation for future empirical studies to investigate learning outcomes and user experience.

VIII. DISCUSSION

The CogTutor prototype demonstrates how cognitive science principles can be incorporated into the design of an AI-assisted learning tool. Instead of providing immediate answers, the system encourages learners to actively retrieve information, explain their reasoning, reflect on their understanding, and then receive feedback. This structured workflow aligns the prototype with established learning strategies while illustrating an alternative approach to traditional AI question-answering systems [1]–[4]. Although the current implementation is a proof of concept, it demonstrates the feasibility of integrating cognitive science into educational AI tools [1], [8].

IX. LIMITATIONS

This project has several limitations. First, the prototype currently operates using predefined instructional content rather than dynamically generated responses. Second, the system was evaluated through functional testing only, and no formal user study was conducted to measure learning outcomes or usability. Finally, the prototype includes a limited number of learning topics and does not yet provide personalized feedback based on learner performance. Future work should evaluate the system with a larger group of learners and incorporate adaptive feedback to improve the learning experience.

X. FUTURE WORK

Future work will focus on expanding CogTutor with adaptive feedback, personalized learning recommendations, and LLM-assisted adaptive content generation using a large language model. A formal user study with students will be conducted to evaluate learning outcomes, engagement, and usability. Additional learning topics, progress tracking, and integration with learning management systems could further improve the system and support long-term learning [1], [4].

XI. CONCLUSION

This paper presented **CogTutor**, a cognitive-science-inspired AI study assistant that encourages active learning through retrieval practice, guided hints, self-explanation, metacognitive reflection, and delayed feedback. The Streamlit-based prototype demonstrates how cognitive science principles can be incorporated into the design of educational AI tools. Although the current implementation is proof of concept, it provides a foundation for future research on AI-assisted learning systems that promote learner engagement rather than passive answer consumption.

ACKNOWLEDGMENT

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Appendix A

Pilot Evaluation Questionnaire

- Rate the following (1=Strongly Disagree, 5=Strongly Agree)
 1. The retrieval question encouraged me to think before receiving help.
 2. The guided hints supported my reasoning.
 3. The self-explanation stage improved my understanding.
 4. The reflection questions helped me monitor my learning.
 5. Delaying the final answer encouraged active learning.
 6. Overall, CogTutor was easy to use.
- Open-ended questions:
 7. Which feature helped you most?
 8. What improvements would you suggest?

Appendix B Project Plan Summary

The project plan was updated throughout development as implementation progressed. All planned milestones were completed successfully. Minor adjustments were made to improve the user interface, refine the evaluation methodology, and incorporate instructor feedback. The total estimated effort for the project was approximately **70 hours**, including literature review, prototype development, testing, documentation, report writing, and presentation preparation.

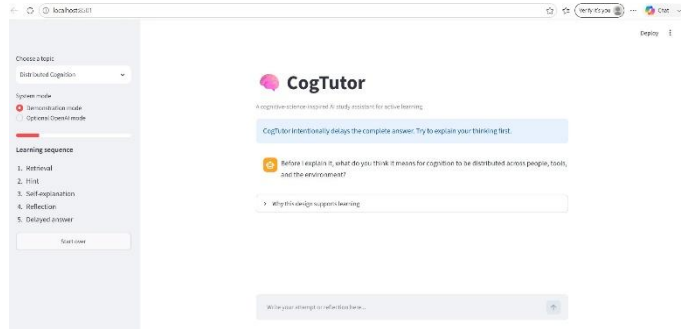
	A	B	C	D	E
1	Week #	Task #	Task Description	Estimated Time (Hours)	Complete? (Y/N)
2	2	1	Create the template task list.	1	Y
3	2	2	Select and refine research question	1	Y
4	2	3	Review cognitive science literature	4	Y
5	2	4	Review AI tutoring systems literature	2	Y
6	2	5	Develop project proposal	2	Y
7	3	6	Prepare and submit project pitch	2	Y
8	3	7	Refine research question from feedback	2	Y
9	3	8	Design overall learning workflow	2	Y
10	3	9	Design application interface	2	Y
11	3	10	Plan Streamlit architecture	2	Y
12	PROJECT PITCH DUE				
13	4	11	Implement home interface	2	Y
14	4	12	Implement retrieval practice stage	2	Y
15	4	13	Implement guided hint stage	2	Y
16	4	14	Implement self-explanation stage	2	Y
17	4	15	Implement metacognitive reflection stage	2	Y
18	5	16	Implement delayed answer stage	1	Y
19	5	17	Implement learning dashboard	2	Y
20	5	18	Debug application and Functional testing	3	Y
21	5	19	Prepare and submit optional midpoint check-in	2	Y
22	OPTIONAL MIDPOINT CHECK-IN DUE				
23	6	20	Prepare midpoint update	2	Y
24	6	21	Create workflow diagram	2	Y
25	6	22	Capture application screenshots	2	Y
26	6	23	Develop evaluation questionnaire	2	Y
27	7	24	Write introduction and background	1	Y
28	7	25	Write implementation section	1	Y
29	7	26	Write evaluation section	1	Y
30	7	27	Write discussion and limitations	1	Y
31	7	28	Write future work and conclusion	2	Y
32	8	29	Format report in IEEE style	1	Y
33	8	30	Proofread and revise report	1	Y
34	8	31	Finalize references and appendices	2	Y
35	8	32	Prepare GitHub repository	2	Y
36	8	33	Upload source code and documentation	2	Y
37	FINAL REPORT DUE				
38	9	34	Prepare presentation slides	2	Y
39	9	35	Practice presentation	2	Y
40	9	36	Revise presentation	1	Y
41	9	37	Final report submission	1	Y
42	9	38	Final presentation submission	1	Y
43	9	39	Final project completion	1	Y
44	FINAL PRESENTATION DUE				
	D	E	F	G	H
Estimated Time (Hours)	Complete? (Y/N)		Total Hours		70
1	Y				
1	Y				
4	Y				
2	Y				

Note: The project plan is organized according to the course week schedule. Several tasks, including report writing,

implementation, documentation, GitHub preparation, and presentation development, were completed progressively throughout the project and often earlier than the final due date. Although the final report and presentation were officially due on **July 20**, many of the activities listed for Weeks 8 and 9 were completed during those course weeks and refined before the final submission.

Appendix C Complete CogTutor User Interface

Appendix D: Code Repository



GitHub Repository

Repository URL: <https://github.com/shabihasetu/CogTutor-AI-Study-Assistant-GT>

Repository Access Note

The project instructions recommend using a private repository or the Georgia Tech GitHub organization. My project repository is hosted on my personal GitHub account rather than the Georgia Tech GitHub organization. To avoid any access issues for the course staff during grading, I have temporarily kept the repository **public** so that the instructor and TAs can access the source code without requiring collaborator invitations or additional permissions. If preferred, I am happy to make the repository private after grading or provide access through an alternative method.